

MVP: Guided design. Customizable. Real products, real budgets, real rooms.

CORE APPROACH

An MVP that guides users through bathroom design customization step by step. Pick categories, describe your vision, set budget and room size, get a coordinated bundle of real KOHLER products — laid out to true scale in the actual room.

- **Extraction Agent (Theme Matching)**
Reads freeform vision text and maps it to one of 25 KOHLER design themes using weighted keyword matching — always confident, never dead-ending.
- **Recommendation Agent (Bundle Building)**
Selects a cohesive set of real products from ~1,500 SKUs that match the chosen theme, stay within budget, and work together as a system.
- **Placement Agent (Spatial Layout)**
Positions each product in the room to true scale, respects door swing clearance, and reports honestly when the room is tight — no silent overlaps.

3

agents working in sequence

~1,500

real KOHLER products

25

design themes supported

0

external LLM calls

Three agents, one vision: Sequential processing pipeline



Each agent is deterministic and explainable. The extraction agent scores keywords; the recommendation engine weighs products by theme match and price; the placement engine uses hard geometric rules. No external API calls. No black boxes.

TECH STACK

Simple, fast, auditable

Backend

Python + Flask

REST API with 5 focused endpoints, no framework overhead.

Frontend

React 18 + Vite

Single-page app with guided conversational UX.

Logic Layer

Deterministic keyword matching + geometry

No external LLM. Fast, reproducible, predictable.

Data

Scraped KOHLER catalog (~1,500 SKUs, real prices)

Real products, real pages, not placeholders.

Visualization

Hand-rolled isometric SVG + 2D plan view

Renders from placement coordinates, never staged.

Testing

Custom Python harness + 1,625-scenario sweep

Placement correctness verified exhaustively.

Tightly bound. Keyword-driven. Thematic. Cost-effective. Scalable.

- **Keyword-matching thematic engine.**
No expensive LLM calls. Deterministic keyword scoring maps user vision to one of 25 KOHLER design themes — fast, reproducible, auditable.
- **Tightly bound product selection.**
Every recommendation is a real KOHLER product with real price and real product page. The bundle you see is the bundle you can actually buy.
- **Geometric honesty in spatial layout.**
Placement engine respects door swing, entrance clearance, and fixture coincidence rules. Reports constraint violations honestly, never hides overlaps.
- **Highly scalable foundation.**
Fixed categories, deterministic logic, and precomputed catalog mean the system scales to thousands of users without needing to retrain or pay per-request fees.

github.com/NikeBhide/NikhilBhideKohler